Parallel and Distributed Computing (CS3006)

Date: October 29, 2026

Sessional-II Exam

Total Time (Hrs.):	1
Total Marks:	45
Total Questions:	3

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Roll No _____ Section _____ Student Signature _____

Instructions

1. Answer **all** questions on the answer sheet.
2. If a question seems ambiguous, state your assumption clearly and solve it.
3. Show your reasoning: **a correct answer without correct reasoning receives no credit.**
4. Write neatly and concisely. Attach this question paper to your answer sheet.

✪ Recall and apply ★ Reasoning and judgment

Question 1 Principles of parallel algorithm design — 15 marks

CLO 3 (C4) — Analyze the performance, scalability, and efficiency of parallel and distributed systems.

An 8×8 sparse matrix A is multiplied by a dense vector b to compute $y = A\,b$. Task i computes $y[i]$ using row i of A, and needs $b[j]$ for every non-zero $A[i][j]$. Task i owns $b[i]$. Figure 1 shows the non-zero pattern of A.

Cost model. Processing each non-zero entry of A costs 1 time unit (one multiply-add). Fetching one remote element of b costs 2 time units, charged to the process that fetches it (the owner pays nothing to send). Each process fetches every distinct remote element it needs exactly once, all processes fetch in parallel, and communication is not overlapped with computation. Two processes, P0 and P1, are used.

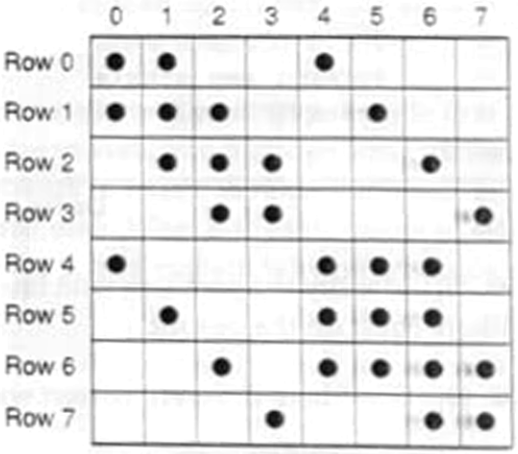


Figure 1. Non-zero pattern of A (● = non-zero entry).

(a) ✪ [3] **Mapping 1 (block of rows).** P0 gets rows 0–3 and P1 gets rows 4–7. Write the lists C_0 and C_1 (the indices of b that each process must fetch from the other process), the total number of fetches, and the computation work on each process.

(b) ✪ [4] **Mapping 2.** P0 gets rows $\{0,1,4,5\}$ and P1 gets rows $\{2,3,6,7\}$. List the edges of the undirected task-interaction graph that this mapping cuts. Write C_0 and C_1, the total number of fetches, and the computation work on each process.

(c) ★ [4] For Mapping 2, are the number of cut edges and the number of fetches equal? Explain why or why not.

(d) ★ [4] **Comparison.** Calculate the efficiency of Mapping 1 and of Mapping 2. The serial time is the time for a single process to handle all non-zero entries of A. Which mapping is better, and why?

Question 2 OpenMP (the common core) — 15 marks

CLO 3 (C4) — Analyze the performance, scalability, and efficiency of parallel and distributed systems.

Listing 1 builds a histogram of the N values in `data`; each `data[i]` is in the range 0 to `BINS`-1.

```
1  #define N 100000000
2  #define BINS 64
3  long hist[BINS] = {0};
4  long total = 0;
5  int *data = malloc(N * sizeof(int));
6  fillData(data, N);  // Assume: 0 <= data[i] < BINS
7  #pragma omp parallel
8  {
9      #pragma omp for
10     for (int i = 0; i < N; i++) {
11         hist[data[i]]++;
12     }
13     #pragma omp for reduction(+:total)
14     for (int b = 0; b < BINS; b++) {
15         total += hist[b];
16     }
17     #pragma omp single
18     printf("Total = %ld\n", total);
19 }
```

Listing 1. Parallel histogram.

(a) ✪ [3] Identify the problem with line 11 of Listing 1. Explain why it occurs and state one likely effect on the output.

(b) ✪ [8] Give three different correct ways to fix the problem you identified in (a).

(c) ★ [4] Is an additional barrier required between the histogram loop (lines 9–12) and the loop that computes `total` (lines 13–16)? Would it be safe to add `nowait` on line 9? Justify both answers.

Question 3 One-to-all broadcast on a combined network — 15 marks

CLO 2 (C6) — Design parallel algorithms and distributed solutions using modern programming models to solve computational problems.

Figure 2 shows a 32-node network made of three sub-networks (an 8-node ring, a 4×4 mesh whose node M(r,c) is labelled rc, and a 3-D hypercube) joined by two bridge links, B1 (R4–M(0,0)) and B2 (M(3,3)–H000). Node **R0** broadcasts an m-word message to all nodes.

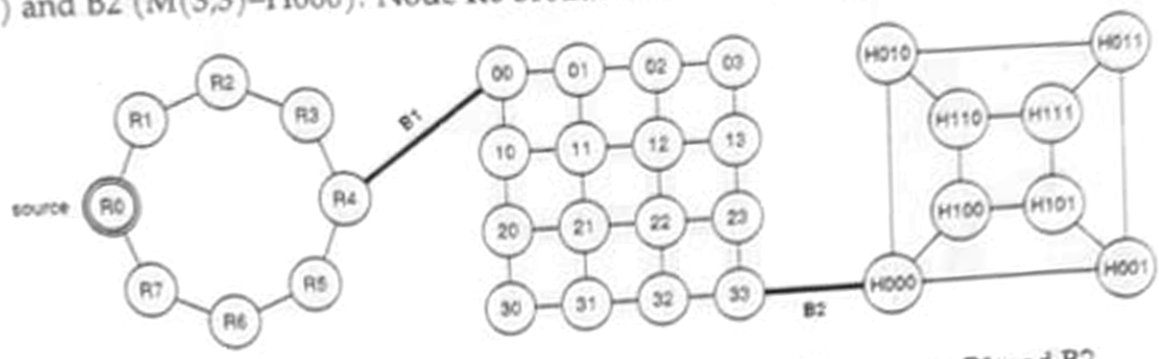


Figure 2. A ring, a 4×4 mesh and a 3-D hypercube joined by bridges B1 and B2.

- Every message has the same start-up time t_s, and every link, **including B1 and B2**, has the same per-word time t_w. Nodes are **single-port**: in each step a node can send one message and, at the same time, receive one message. A message costs $t_s + t_w m$ however many links it crosses (cut-through routing; the per-hop time t_h is ignored). The message is always sent whole (never split into pieces), and steps are numbered from 1.
- Bridge nodes are gateways: a message crosses a bridge only as a send from one bridge node to the other; it cannot be routed *through* a bridge.

(a) ✪ [5] **Strategy S1:** each sub-network broadcasts by recursive doubling from the node where the message enters it, and a bridge node forwards the message across its bridge only after its own sub-network has finished. By the end of which step has each of R4, M(0,0), M(3,3) and H000 received the message? What is the total cost of S1?

(b) ★ [10] Now abandon S1 and use **Strategy S2** for this part: in each step, a node that has the message may send it to *any* node of its own sub-network, and a bridge node may send it across its bridge without waiting for its sub-network to finish. A node that receives the message in step x can send it in step $x + 1$ or any later step. What is the **minimum possible cost** of the broadcast? Give a schedule that achieves it, and explain why no schedule can do better.

— End of Exam —